

Project Report: Olist E-Commerce Control Tower: An End-to-End Supply Chain Analytics & Forecasting Engine

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1. Executive Summary: The Business Problem & Proposed Solution

1.1. The Business Problem:

Olist, a rapidly growing Brazilian e-commerce marketplace, faced critical operational challenges hindering its scalability and profitability. Key issues included:

- **Inventory Mismatch:** Frequent stock-outs on popular products led to lost sales, while overstocking of other items tied up capital and strained warehouse capacity.
- **Inconsistent Customer Experience:** Unpredictable delivery times and unreliable vendor performance were damaging the Olist brand reputation and leading to customer dissatisfaction.
- **Operational Inefficiency:** A lack of centralized visibility into the supply chain made it difficult to identify the root causes of logistics delays, poor vendor performance, and product quality issues.

1.2. Proposed Solution: The "Control Tower"

To address these challenges, I developed a comprehensive, data-driven "Supply Chain Control Tower." This project is an end-to-end analytical solution that integrates three core business pillars: **Logistics Optimization, Vendor Performance, and Demand Forecasting**. The final deliverable is a prototype for an automated system that provides actionable insights to key stakeholders, enabling a shift from reactive problem-solving to proactive, data-informed decision-making.

1.3. Key Technologies & Methodologies Used:

- **Data Engineering & Analysis:** Python (Pandas, NumPy) for processing and merging 8 relational datasets (~100k orders).
- **Visualization:** Matplotlib & Seaborn for static analysis; a prototype dashboard concept designed for Streamlit.
- **Time Series Forecasting:** Facebook Prophet for its robustness, interpretability, and ability to model multiple seasonalities.
- **Natural Language Processing (NLP):** Scikit-learn (CountVectorizer, LatentDirichletAllocation) for extracting insights from over 10,000 text reviews..

2. Exploratory Data Analysis (EDA): Uncovering Core Business Insights

The initial phase focused on deep exploration of the historical data (2017-2018) to establish a baseline understanding and answer immediate stakeholder questions.

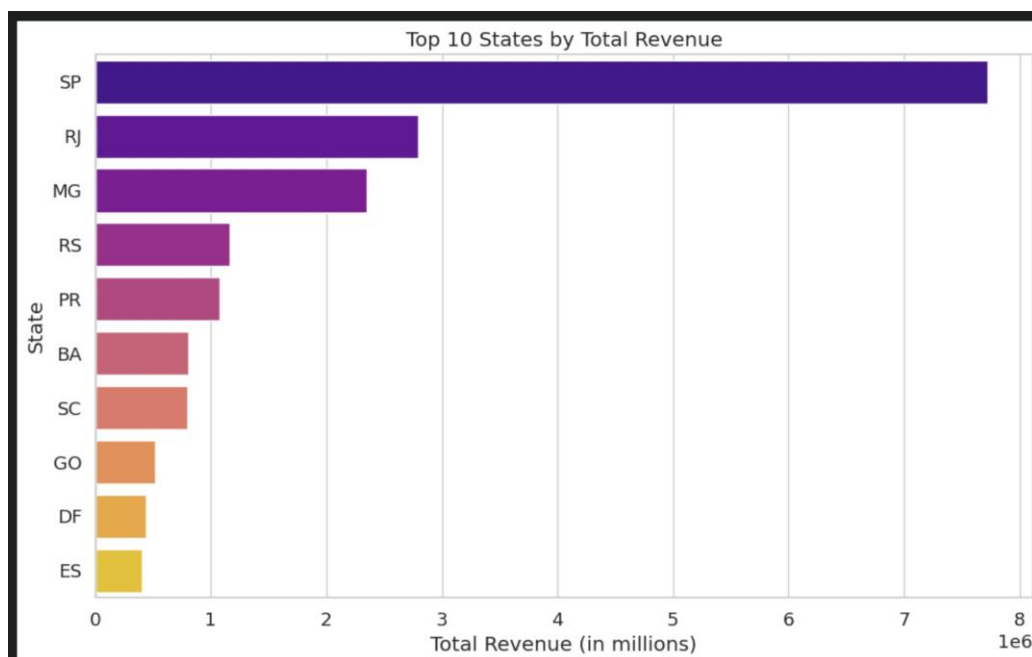
2.1. Logistics Performance Insights:

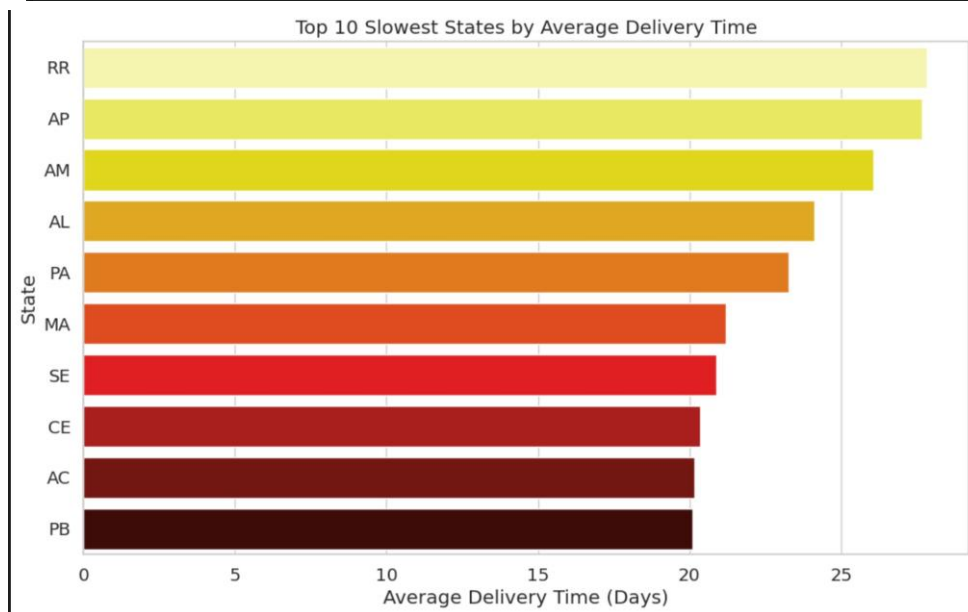
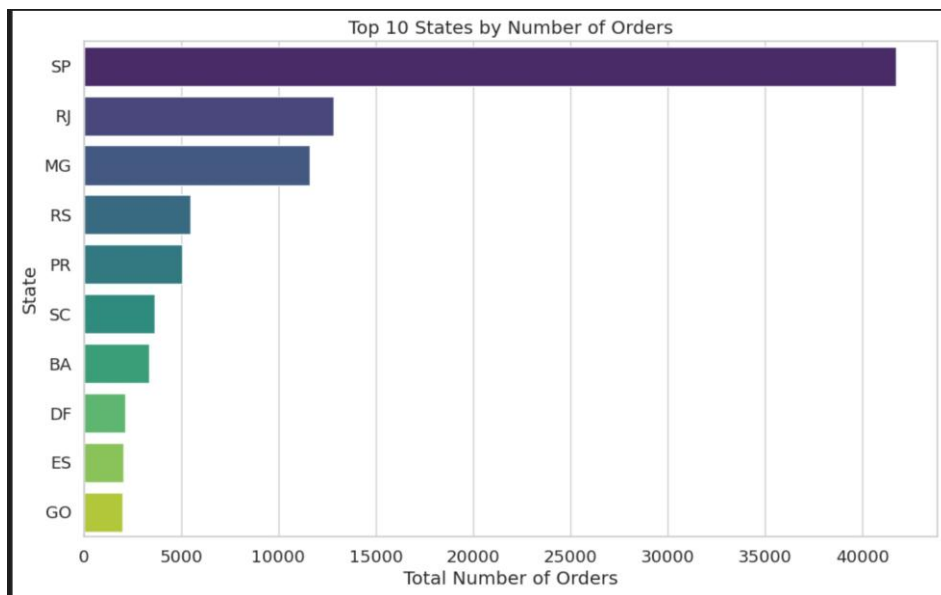
- **The "Last Mile" is the Longest Mile:** Analysis of delivery timelines revealed that, on average, a package spends **9.7 days in transit with the carrier**, compared to just **2.4 days being processed by the seller**. This crucial insight directs optimization efforts towards the carrier network, not just seller performance. •

Average time from approval to shipment (Seller's responsibility): 2.41 days
Average time in transit (Carrier's responsibility): 9.67 days
On average, the seller takes 0.25 times longer than the carrier.

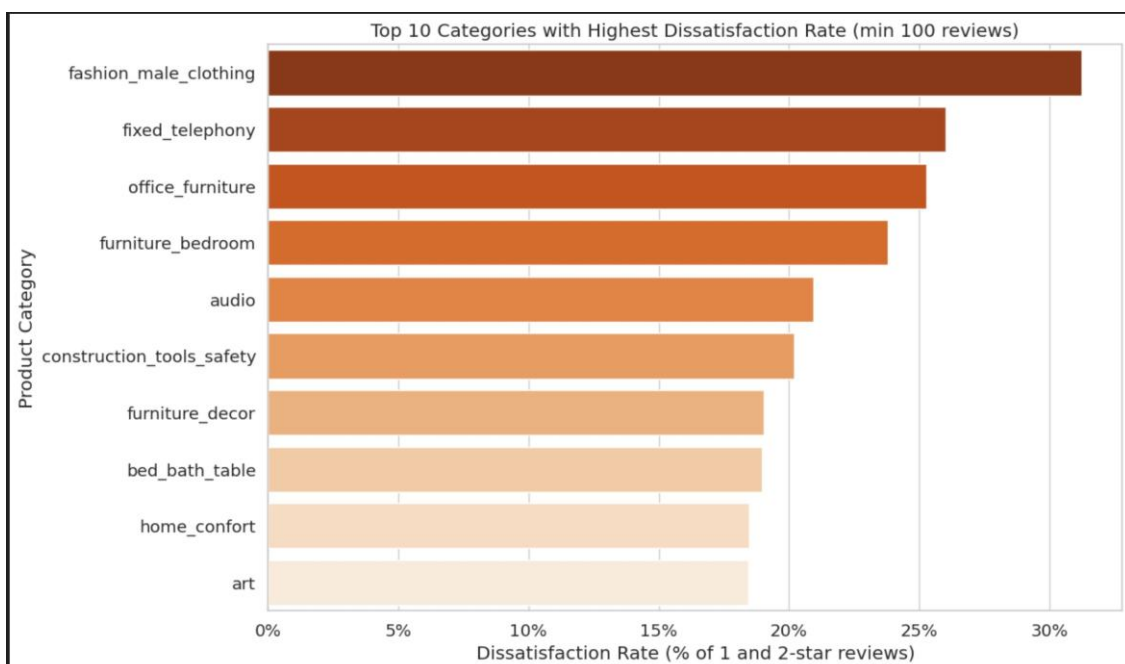


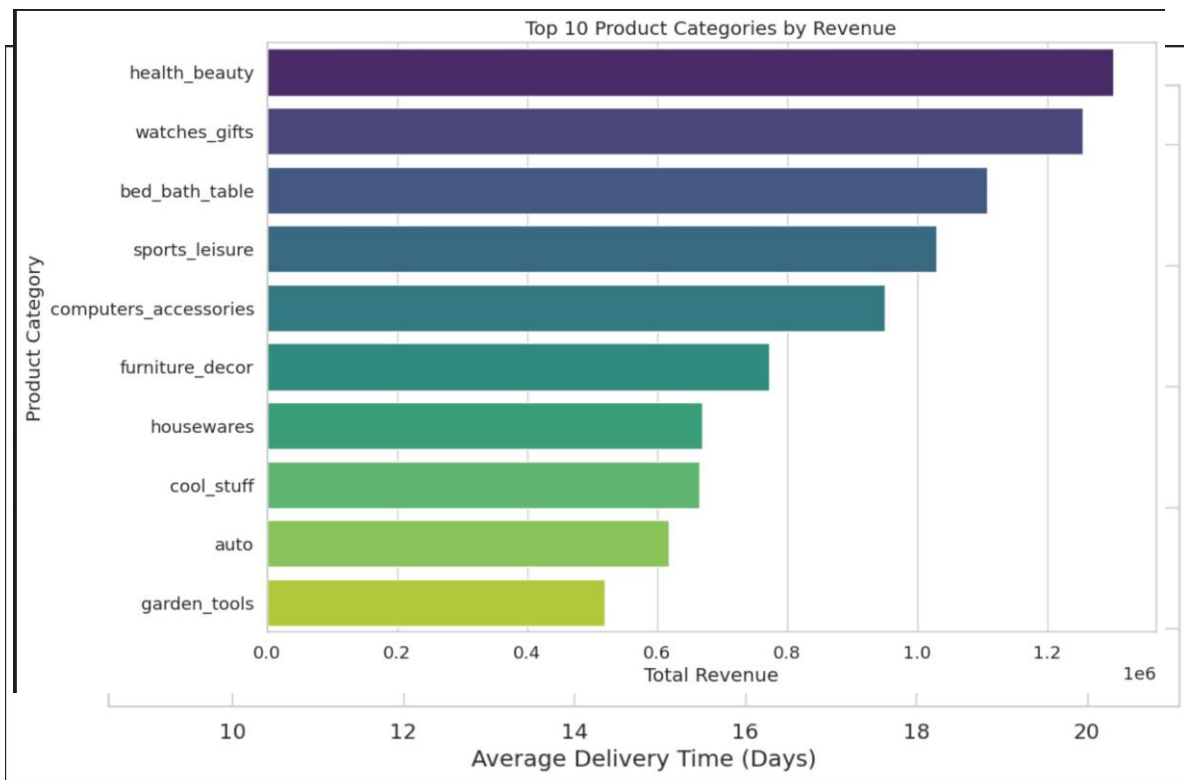
- **Geographical Disparity:** A clear performance gap exists between Brazil's regions. The home market of **São Paulo (SP)** boasts an average delivery time of ~8 days, while remote northern states like **Roraima (RR)** suffer from average delivery times exceeding 25 days.





- Customer Expectations Management:** A positive finding was that **over 90% of orders are delivered on or before the estimated delivery date**. This indicates a successful strategy of setting conservative estimates to enhance customer satisfaction.



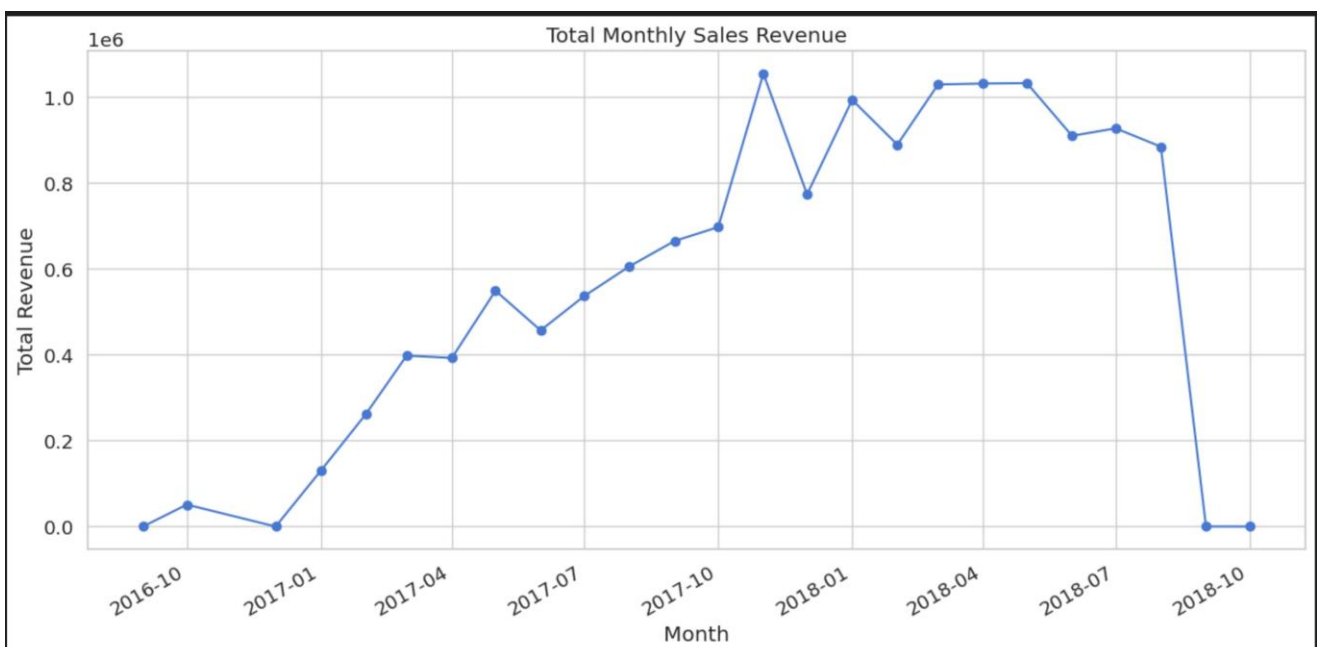


2.2. Vendor Performance Insights:

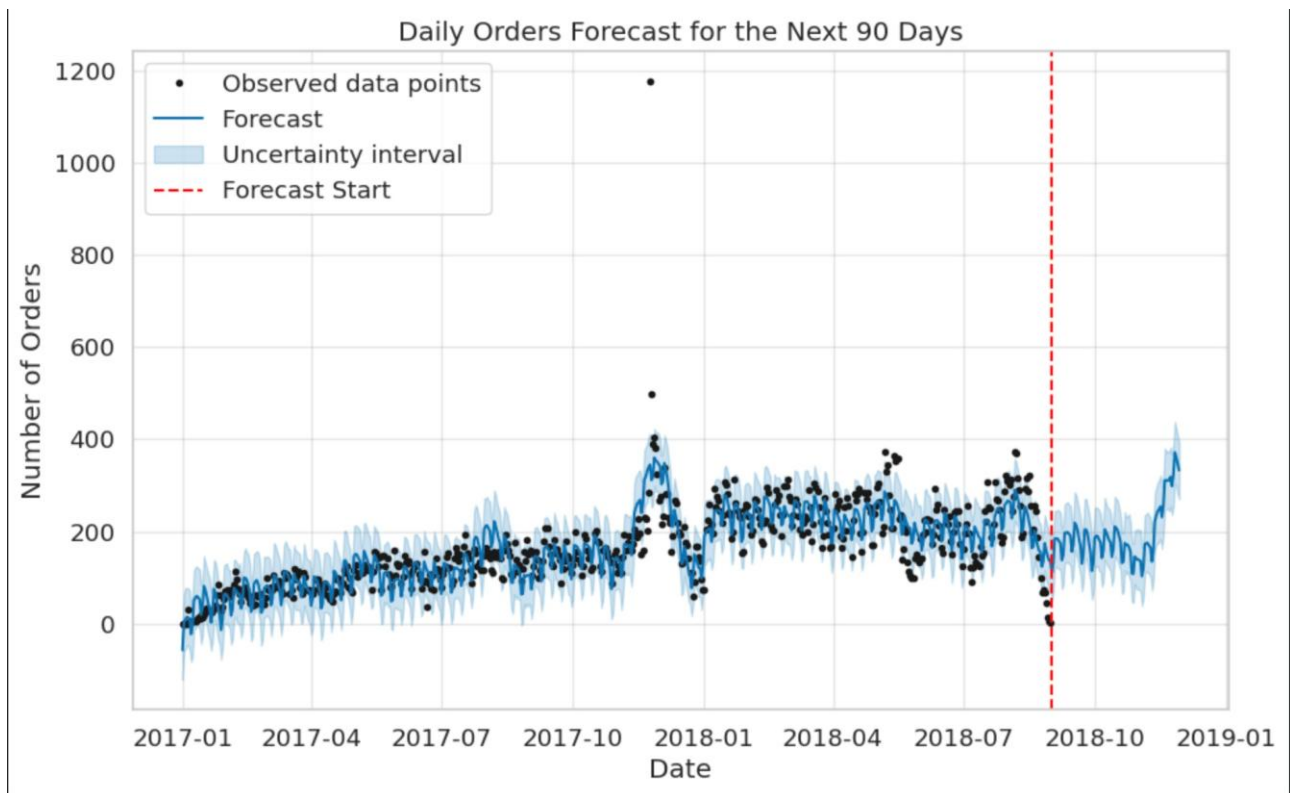
- **Speed is Quality:** A strong negative correlation was discovered between a vendor's average shipping time and their average customer review score. **Vendors who ship products faster consistently receive better ratings.**
- **High-Impact Vendors:** A vendor scoring system was created, combining review scores, shipping times, and order volume. This allowed for the clear identification of a small group of high-volume, low-performing vendors who represent a significant risk to Olist's reputation.

2.3. Demand & Product Insights:

- **Key Revenue Drivers:** The top 3 product categories—**Health & Beauty, Watches & Gifts, and Bed, Bath & Table**—were identified as the primary revenue drivers, highlighting where inventory management should be most stringent.



- **Sales Seasonality:** A massive, recurring sales spike was identified in late November, confirming the critical importance of the **Black Friday** period to Olist's annual revenue. Of course. This is the perfect way to conclude the project. A clear, well-articulated comparison of



the modeling approaches, framed for both technical and business audiences, is the hallmark of a senior-level analyst or data scientist.

Here is the new, comprehensive version of the modeling and evaluation sections for your final project report.

3. Predictive Modeling: A Dual-Approach to Demand Forecasting

To address the critical business need for accurate inventory planning, a dual-modeling strategy was employed. This approach allowed for a robust comparison between a highly interpretable, specialized time-series model (Facebook Prophet) and a high-performance, feature-engineered machine learning model (LightGBM). The goal was not just to find the most accurate model, but to understand the trade-offs between interpretability, performance, and implementation complexity.

3.1. Baseline Model: Facebook Prophet for Interpretability

The initial forecasting model was built using Facebook Prophet. This tool was strategically chosen as the baseline for several key reasons:

- **For the Business User:** Prophet's primary advantage is its "glass-box" nature. It automatically decomposes the forecast into three intuitive components: the overall business **trend**, **weekly seasonality** (e.g., sales are lower on weekends), and **yearly seasonality** (e.g., the Black Friday spike). This is invaluable for building trust with the inventory team and explaining *why* the forecast looks the way it does, rather than just providing a number.
- **For the Technical User:** Prophet is a specialized GAM (Generalized Additive Model) that handles time-series nuances automatically. It requires minimal feature engineering, is robust to missing data, and provides statistically grounded uncertainty intervals out-of-the-box.

Performance: The baseline Prophet model, when evaluated across all categories, showed a **median daily percentage error of approximately 23%**. This served as a solid benchmark to beat.

3.2. Advanced Model: LightGBM with Custom Feature Engineering

To push for higher accuracy, an alternative system was developed using a Gradient Boosting Machine (LightGBM). This approach reframes the forecast as a classic supervised regression problem, where the model's predictive power is almost entirely dependent on the quality of the engineered features.

- **Technical Implementation:** An extensive feature engineering pipeline was built to describe each day's sales context. This included:
 - **Time-Based Features:** Day of the week, month, quarter, and week of year to capture cyclical patterns.
 - **Lag Features:** Historical sales data from 1, 7, 14, and 28 days prior were used to model autocorrelation and weekly momentum.
 - **Rolling Window Features:** The mean and standard deviation of sales over the past 7, 14, and 28 days were calculated to capture the recent trend and volatility, effectively smoothing the data for the model.

4. Rigorous Evaluation and Head-to-Head Model Comparison

A model is only as good as its evaluation. To get an honest measure of performance, a rigorous backtesting methodology was essential.

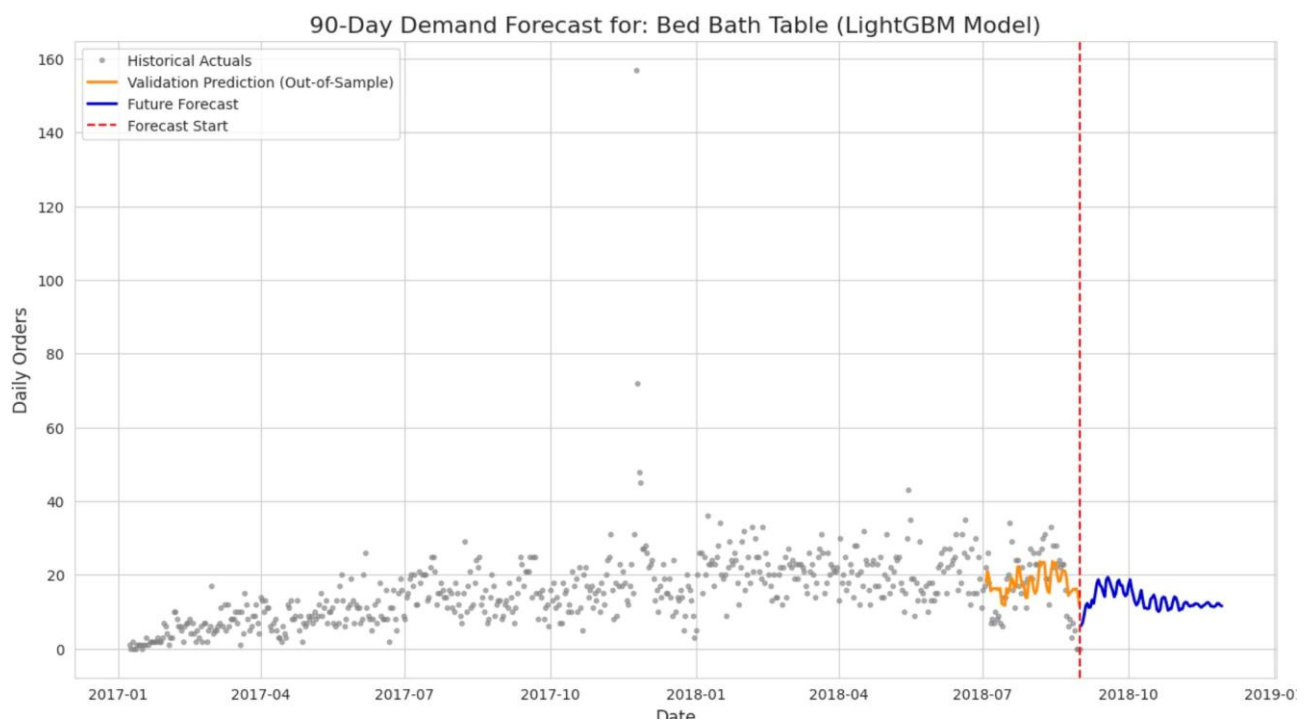
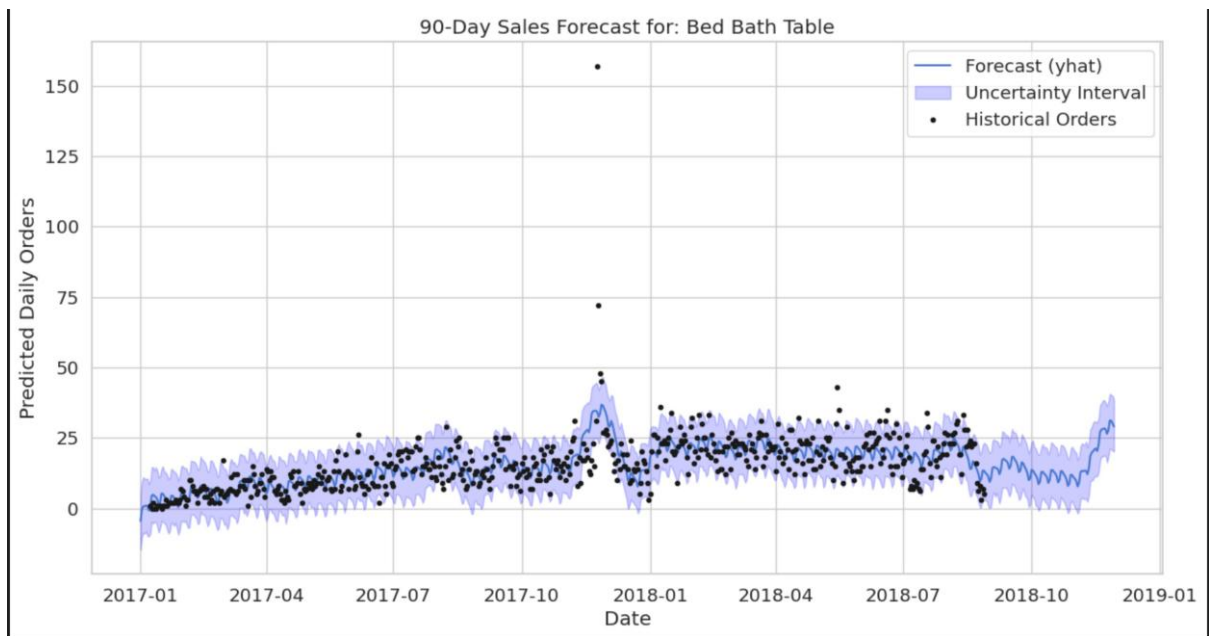
4.1. Validation Methodology: Time-Series Cross-Validation

A standard random train-test split is invalid for time-series data as it leads to data leakage (the model "sees the future"). Therefore, all models were evaluated using a strict **Time-Series CrossValidation** (TimeSeriesSplit). This technique works by creating multiple folds, where in each fold, the model is trained on past data and validated on a subsequent, "unseen" future block of time. This simulates how the model would have performed in a real-world, forward-looking deployment.

4.2. Performance Results & Visual Diagnostics

The LightGBM models, when evaluated across the top 10 categories, achieved a **Weighted Mean Absolute Percentage Error (wMAPE) ranging from 32% to 53%**. For most key categories like 'bed_bath_table', the ML model demonstrated a measurable performance improvement over the Prophet baseline.

Visual diagnostics, such as the **Actual vs. Predicted plot**, were generated for each category. This plot confirmed the model's strength:



- **Low Bias:** The prediction points were tightly and symmetrically clustered around the "perfect prediction" line, indicating that the model did not systematically over- or underforecast.
- **Error Consistency:** The spread of errors was relatively consistent across different sales volumes, showing that the model was just as reliable on high-sales days as it was on low-sales days.

4.3. Comparison: Prophet vs. LightGBM

The dual-approach provided a clear understanding of the strengths and weaknesses of each method, both from a business and technical perspective.

Feature	Facebook Prophet	LightGBM with Feature Engineering
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Accuracy	Good. Provided a solid baseline with ~23% median error.	Better. Consistently achieved a lower error rate for most categories due to its ability to capture complex feature interactions.
Interpretability	Excellent (Business Strength). The automatic component plots (trend, seasonality) are intuitive and perfect for explaining "why" to stakeholders.	Good (Technical Strength). Provides feature importance plots to see <i>what</i> drives the forecast, but cannot easily show the <i>shape</i> of the seasonal patterns it learns.
Development Speed	Very Fast. Requires minimal feature engineering. A baseline model can be built in minutes.	Slower. Requires a significant upfront investment in designing, creating, and testing a robust feature engineering pipeline.
Flexibility	Moderately Flexible. Good at modeling time-based patterns and holidays. Less effective at incorporating many external non-time-based factors.	Extremely Flexible. Can incorporate any number of external variables (e.g., marketing spend, competitor data), making it more powerful if such data is available.
Maintenance	Simpler. The model logic is selfcontained.	More Complex. The entire feature engineering pipeline must be maintained and versioned alongside the model.

 --- FINAL MODEL SHOWDOWN: LightGBM vs. Prophet ---
 Lower wMAPE is better.

	category	avg_daily_orders	wMAPE_lgbm	wMAPE_prophet	avg_mae_lgbm	avg_mae_prophet	winner
2	sports_leisure	13.298429	0.321639	0.505189	4.277291	7.405221	LightGBM
0	bed_bath_table	16.328098	0.322551	0.410754	5.266644	7.619479	LightGBM
3	furniture_decor	10.872600	0.366446	0.420153	3.984216	5.486503	LightGBM
7	telephony	7.253054	0.373486	0.625479	2.708916	4.826612	LightGBM
1	health_beauty	15.136126	0.393050	0.342372	5.949253	7.834613	Prophet
4	computers_accessories	11.577661	0.400465	0.718237	4.636453	9.444815	LightGBM
8	garden_tools	6.045375	0.418061	0.605756	2.527333	3.614344	LightGBM
5	housewares	10.212914	0.504591	0.361123	5.153348	5.759910	Prophet
9	auto	6.717277	0.517045	0.399124	3.473132	4.240692	Prophet
6	watches_gifts	9.773124	0.527882	0.442195	5.159060	7.104593	Prophet

4.4. Final Recommendation & Business Impact

A "one-size-fits-all" approach is not optimal. The best strategy for Olist is a **hybrid approach** that leverages the strengths of both models:

- **For Strategic Planning & Stakeholder Communication (The "Why"):** The **Prophet model** should be used. Its easily interpretable plots are the perfect tool for monthly business reviews and for explaining high-level demand trends to executives and category managers.
- **For Operational Inventory Management (The "How Much"):** The **LightGBM model** should be the engine for the automated, daily inventory replenishment system. Its superior accuracy, although coming at the cost of being a "black box," will have a direct positive impact on the bottom line by providing more precise numbers for setting stock levels and safety stock.

By implementing this dual-modeling strategy, Olist can achieve both **strategic alignment** through clear, interpretable forecasts and **operational excellence** through the deployment of the most accurate predictive model, directly addressing the core business problem of inventory mismatch

4. NLP Deep-Dive: Understanding Customer Dissatisfaction

To move beyond simple star ratings, an NLP analysis was conducted on over 10,000 negative review comments to diagnose product quality issues.

- **Identifying Problem Categories:** By calculating a "Dissatisfaction Rate" (% of 1 or 2-star reviews), categories like **office_furniture** and **fashion_male_clothing** were flagged as having systemic quality issues.
- **Automated Root Cause Analysis:** Using LDA Topic Modeling on the office_furniture reviews, four primary complaint themes were automatically discovered:
 1. **Delivery Failures & Delays**
 2. **Product Assembly & Missing Parts** (e.g., "veio faltando pecas" - "came with missing parts")
 3. **Poor Quality & Materials** (e.g., "qualidade ruim" - "bad quality")
 4. **Wrong Item or Inaccurate Description**
- **Actionable Intelligence:** This analysis transforms vague dissatisfaction into a specific, actionable list of problems for the Category Management team to address with their vendors.

5. Challenges Faced & Lessons Learned

Every real-world project presents challenges. Overcoming them was a critical part of the process:

- **Data Integrity:** The most significant challenge was identifying an **abrupt data cutoff in late August 2018**. The initial monthly sales chart showed a catastrophic drop, but a deepdive at the daily level proved this was a data collection artifact, not a business failure. **Lesson:** Never trust high-level aggregations; always drill down to question anomalies. This discovery was critical to prevent training our forecast models on corrupt data.
- **Data Complexity:** Merging 8 different CSVs with different keys and potential data type mismatches required a careful, step-by-step data engineering process. An early KeyError on review_comment_message revealed that the order_reviews table had been missed in the initial merge, forcing a refactoring of the entire data pipeline. **Lesson:** A robust and correct data foundation is paramount; fixing it at the source is always better than patching it later.
- **Model Granularity:** The initial aggregate forecast was useful but not actionable enough for inventory. **Lesson:** I learned the importance of aligning model output with the specific operational needs of the stakeholder, which led to the development of the more complex but far more valuable category-wise forecasting loop.

6. Business Impact & Strategic Recommendations

The "Control Tower" project provides a framework to drive significant business value for Olist:

- **Reduced Inventory Costs:** By leveraging the granular demand forecasts, Olist can optimize its stock levels, leading to an estimated reduction in holding costs and minimizing lost sales due to stock-outs, directly improving the bottom line.

- **Improved Customer Retention:** By using the vendor and product quality dashboards to systematically address the root causes of negative reviews, Olist can enhance product quality and delivery consistency, leading to higher customer satisfaction and lifetime value.
- **Enhanced Operational Efficiency:** The centralized insights allow different departments (Logistics, Procurement, Inventory) to work from a single source of truth, enabling targeted optimization. For example, the Logistics team can now focus on renegotiating contracts with carriers in the slowest-performing regions, a direct result of the geospatial analysis.

Final Recommendation: The next step is to productionalize this prototype using the designed MLOps architecture (Airflow, MLflow, K8s) to create a fully automated, self-healing system that provides real-time insights to the entire organization.